

SOFTWARE-DEFINED NETWORKS

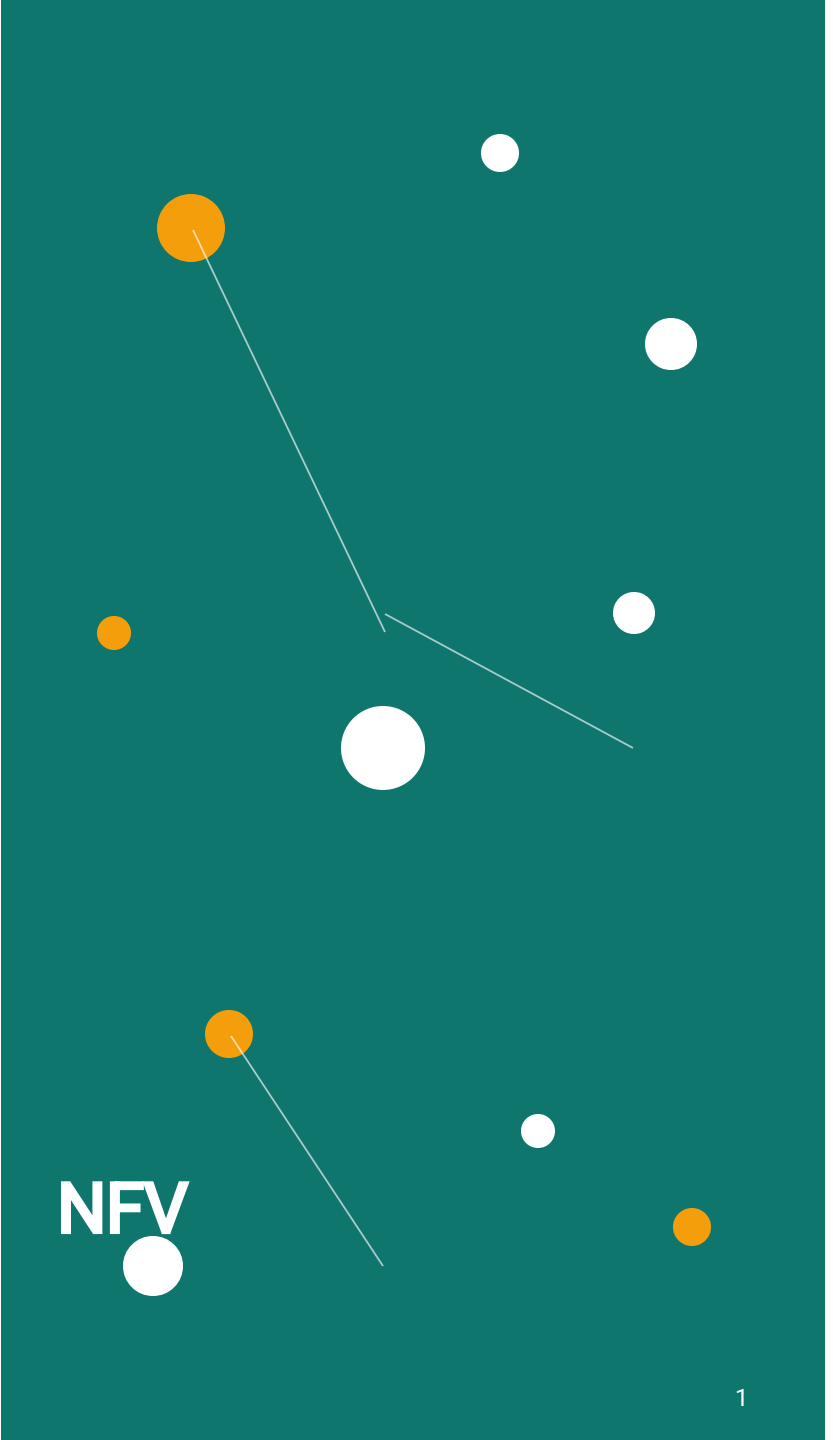
Network Function Virtualization (NFV)

and the OSI Architecture

How virtualized software functions replace dedicated hardware appliances across the OSI stack – decoupling network intelligence from physical boxes.

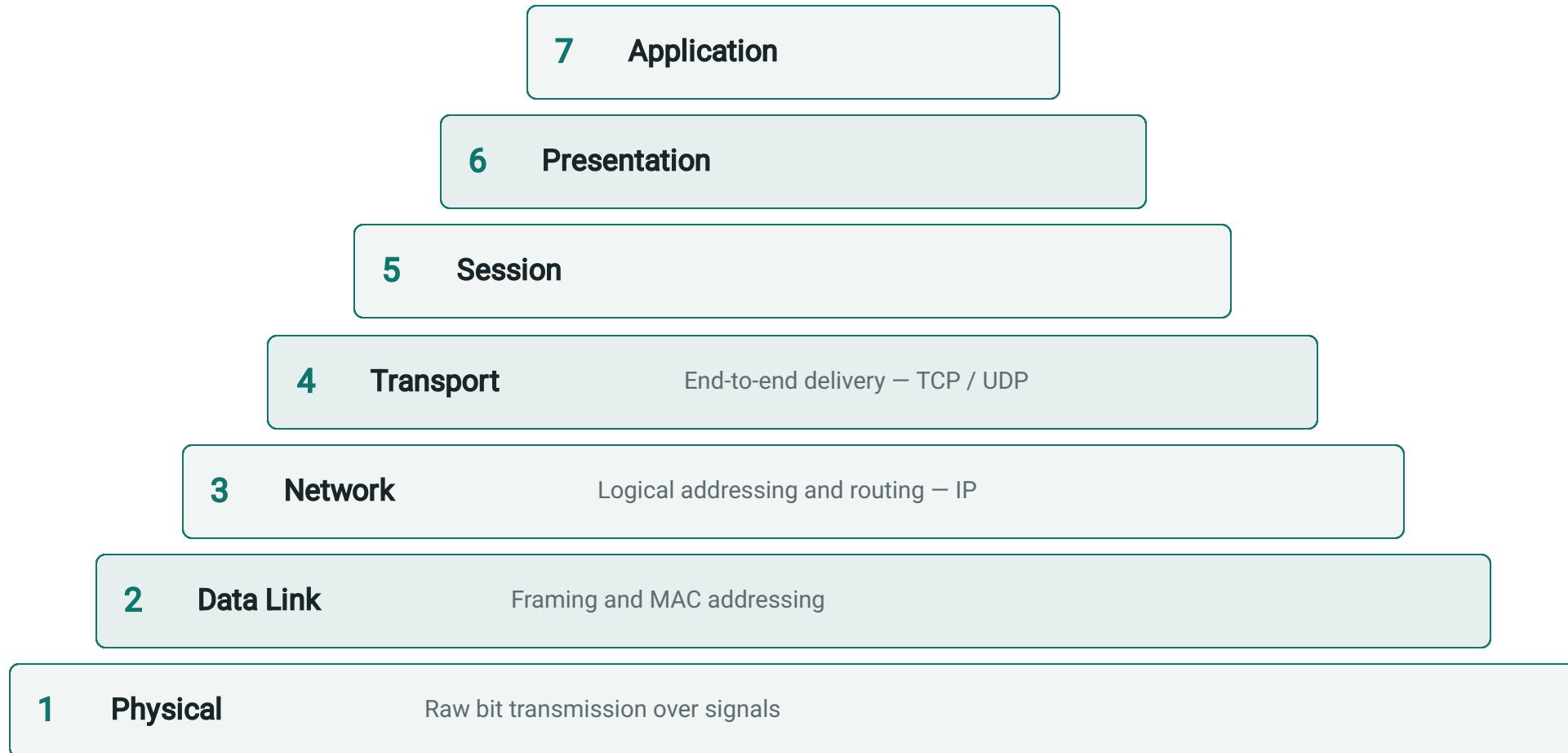
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NFV



The OSI Reference Model

Seven standardized layers, built up from the physical foundation to the application



Layers 5–7 shown with name only above – see full functions in the mapping table ahead.

What Is NFV? The ETSI Framework

NFV replaces purpose-built hardware with software running on standard servers



Virtualized Network Functions

Software versions of routers, firewalls, load balancers, and core-network elements that once lived in dedicated hardware boxes.



NFV Infrastructure

The pooled compute, storage, and network resources — virtualized via hypervisors or containers — that VNFs are deployed on.



Management & Orchestration

The NFV Orchestrator, VNF Manager, and Virtualized Infrastructure Manager coordinate lifecycle, scaling, and placement.

Where NFV Sits Relative to OSI

NFVI virtualizes the lower layers; VNFs implement network and higher-layer functions in software

OSI STACK

7 Application

6 Presentation

5 Session

4 Transport

3 Network

2 Data Link

1 Physical

VIRTUALIZED NETWORK FUNCTIONS (VNFs)

Software instances replace dedicated appliances at Network through Application layers – vRouter, vFirewall, vLoadBalancer, vEPC, vIMS run as VMs or containers.

Elastic · Software-defined · Deployed in minutes

NFV INFRASTRUCTURE (NFVI)

Hypervisors, virtual switches, and commodity servers virtualize physical links and framing – the pooled resource pool VNFs run on.

How NFV Changes the Network

OSI still describes the functions — NFV changes how and where they run

1

Hardware Decoupling

Functions run as software on commodity servers, not proprietary appliances.

2

Elastic Scaling

VNF instances spin up or down automatically to match traffic demand.

3

Service Function Chaining

Traffic is steered through an ordered sequence of virtual functions.

4

SDN Integration

Virtualized functions pair with programmable, centralized control planes.

5

Faster Deployment

New services launch in minutes via orchestration templates.

6

Multi-Tenancy & Isolation

Shared infrastructure hosts isolated functions for multiple tenants.

Layer-by-Layer: Traditional vs. NFV

What ran on dedicated hardware at each OSI layer, and its virtualized equivalent today

OSI LAYER	TRADITIONAL IMPLEMENTATION	NFV EQUIVALENT	EXAMPLE VNF
7 Application	Dedicated app-delivery appliances	Virtualized application functions	vIMS, vCDN
6 Presentation	Hardware encryption/transcoding units	Software crypto & media functions	Virtual transcoder
5 Session	Session border controllers (SBC)	Software session control	vSBC
4 Transport	Hardware load balancers/firewalls	Software transport functions	vLoadBalancer, vFirewall
3 Network	Physical routers, core network boxes	Software routing & core functions	vRouter, vEPC
2 Data Link	Physical switches, dedicated NICs	Virtual switching	vSwitch, SR-IOV
1 Physical	Fixed physical infrastructure	Pooled virtualized infrastructure	NFVI (hypervisor/host)

Conclusion & Outlook

NFV keeps the OSI model's logical separation of functions intact, but frees every function above the physical layer from its own dedicated box — turning the network into software that can be deployed, scaled, and chained on demand.

OSI still defines what each function does — NFV changes only where and how it runs

MANO orchestration turns service rollout from a hardware project into a software deployment

NFV pairs naturally with SDN and underpins virtualized 5G/6G core networks

Minutes

VS. WEEKS TO DEPLOY

70%+

POTENTIAL CAPEX REDUCTION

Native

IN VIRTUALIZED 5G/6G CORES